

Assignment # 2

You are to consider all of the scenarios below. For each scenario, provide an answer to the scenario questions [CLO 2: Discuss Intellectual property rights for open-source software] <Understanding>

[Marks: 10+10=20]

Case Study: The HealthLink Project

HealthLink Technologies is developing a **mobile application for remote patient monitoring**. The app records patients' vital signs using wearable devices and automatically alerts doctors when readings cross critical thresholds. The system is designed to improve hospital efficiency and save lives by detecting emergencies early.

The situation

Sara, a senior software engineer, discovers that the app's data encryption module occasionally fails during large uploads. This could allow unauthorized interception of sensitive medical data. When she reports the issue, management decides to delay fixing it because the upcoming investor demo is crucial and "no real patient will be harmed during trials." Sara feels uneasy but is told she is being "too rigid" about minor issues.

Meanwhile, another team member, Amir, adds a new "wellness insights" feature that uses patient data to recommend lifestyle products from partnered companies. He doesn't inform users that this data is shared with advertisers, arguing that "it's anonymized and everyone benefits."

A few weeks later, the hospital IT administrator, Ms. Farah, requests full access to all patient data to "study system usage patterns." Sara objects that this violates the privacy policy the patients agreed to, but her supervisor approves the request anyway.

During testing, an intern uploads dummy patient data to simulate emergencies but accidentally leaves one file publicly accessible on a shared cloud folder. The development team debates whether they should report this incident to the hospital, fearing reputational damage and potential loss of funding.

Finally, in a design review, Sara notices that the app interface requires multiple taps to confirm medical alerts. This delays nurses' response times. When she suggests simplifying it, her project manager dismisses the concern, saying "our job is code, not usability—let the nurses adapt."

- a. Which ACM Code clause applies to management's decision to postpone fixing the encryption bug? Explain why. [2]
- b. Which ethical principle governs Amir's decision to share user data with advertisers without consent? [2]
- c. Which clause should guide the decision about granting full access to the hospital administrator? [2]
- d. Which ethical responsibility applies to the team's debate about whether to disclose the accidental data exposure? [2]
- e. Which ACM Code principle addresses the neglect of usability concerns that could affect patient safety? [2]

Case Study: MetroRide – The Smart Mobility Startup

MetroRide is a growing ride-sharing startup that uses AI to optimize routes, predict high-demand zones, and allocate drivers efficiently. Its app collects real-time GPS data from both riders and drivers. The company's mission is to make urban transport faster, cheaper, and environmentally friendly.

The situation

Ali, the lead data engineer, discovers that the system's algorithm tends to prioritize wealthier neighborhoods, where data is more complete and demand patterns are stable. As a result, poorer districts have longer wait times and higher cancellation rates. When he raises the issue, the product manager replies, "That's just the data — we can't fix social inequality."

Meanwhile, MetroRide's marketing head, Sana, proposes a new feature called "Driver Efficiency Score." The score uses driver data (average speed, customer rating, idle time) to assign performance bonuses. However, she instructs her team to hide the scoring formula from drivers to prevent "gaming the system." Some drivers complain they are being unfairly penalized without explanation.

During a weekend update, a developer named Hamza accidentally pushes live code that logs passengers' exact trip coordinates to a public analytics dashboard. The issue is quickly noticed but not reported, and the dashboard remains accessible for several hours before being fixed. Management decides not to inform the public, citing that "no names were shown, so no real harm occurred."

Later, Ali learns that MetroRide's AI team has begun using anonymized trip data to develop a predictive policing model for a city law enforcement agency. He worries this might enable surveillance and discrimination against certain communities, but his team lead insists that "we're just providing technology — how it's used is not our concern."

Finally, during a product showcase, Sana claims that the platform is "100% privacy compliant and bias-free," despite knowing that bias issues remain unresolved and some drivers' consent was not properly logged in the system.

- a. Which ACM Code principle applies to the decision to ignore bias affecting certain neighborhoods, and why? [2]
- b. Which ethical clause governs the decision to hide the "Driver Efficiency Score" formula from drivers? [2]
- c. Which ACM Code clause applies to the handling of the unreported dashboard incident? [2]
- d. Which principle should guide Ali's concern about using mobility data for predictive policing? [2]
- e. Which ethical responsibility applies to Sana's public statement about "100% privacy compliance"? [2]